

# The Magic of Half!

## Unlocking Binary Search





Guess the price  
of a cricket bat  
hidden in a  
sorted list of  
prices!



**Secret**

## Price Tag Game

The prices are arranged from lowest to highest. One price has a secret question mark! Can you figure out the hidden price by guessing smartly? Think carefully — which half should you look in?

*Can you find it?* 🔍



# What Is Linear Search?

**1 by 1**

Check every item from the start until you find it!

Linear search means checking every price from the start until we find the secret one. We go through the list one item at a time — first, second, third... all the way until we find what we are looking for! It works but can take a long time if the list is long!





# What Is Binary Search?



## How Binary Search Works

Binary search means looking in the middle, then choosing the left or right half to search next. Each step removes half the remaining items!

**We only look at half the list each time!**

Start in the middle → Too high? Go left! Too low? Go right! → Repeat until found. Fast and smart!



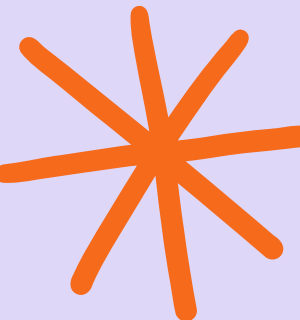
✓ Sorted List  
1, 3, 7, 12, 18, 25  
Numbers in order —  
Yes!

2 5 8 12 12 15 20

## Why Must the List Be Sorted?

Binary search needs the list to be sorted to decide which half to search. Without order, we cannot know if the answer is on the left or the right!

✗ *Jumbled — No!*



# Halving Activity: Let's Try Together!



01

## Steps 1 & 2: List and Middle

Show the entire list of prices. Split it into two halves and highlight the middle element. Is your number bigger or smaller than the middle? Each time, we cut the list in half. Magic!

02

## Steps 3 & 4: Pick & Repeat!

Pick the half where your number could be hiding — and throw away the other half! Keep repeating until you find it. Fewer steps, faster answer!



# How Binary Search Works —

## Simple Steps

01

### **Find the middle item. Stop if found!**

Look at the middle item of the sorted list. Is it the number you are searching for? If yes — you found it, stop! If not, decide which half to explore next based on whether your item is smaller or bigger.

02

### **Search left or right half. Repeat!**

If your item is smaller than the middle, search only the left half. If your item is bigger, search only the right half. Keep halving until you find it — each step cuts the work in half!



# Sorted Lists Around Us — Part 1

Binary search can help in many everyday situations! Sorted lists are all around us — once you spot them, you can use binary search to find things super fast!



**School Assembly**



| Roll Call |         |         |
|-----------|---------|---------|
| 1         | Niec    | 229     |
| 2         | Ysian   | 284     |
| 3         | Gelld   | 386     |
| 4         | Fileres | 252     |
| 5         | Natone  | 258     |
| 6         | Haite   | 253     |
| 5         | Laal    | 251     |
| 6         | Gami    | 222     |
| 1         | Fierico | 116     |
| 2         | Herlyan | 148     |
| 8         | Uklates | 154     |
| 6         | Satxee  | 127     |
| 7         | Salore  | 159     |
| 8         | Poidan  | 1121617 |
| 10        | Fedlea  | 147     |
|           | Sorcer  | 185     |

**Class Roll Numbers**



**Dictionary Words**



# Sorted Lists Around Us — Part 2!

More examples from our daily life! Sorted lists are everywhere around us in India — from the kirana shop to cricket stadiums and our favourite textbooks. Binary search works on all of them!



**Kirana Spice Jars**



**IPL Jersey Numbers**



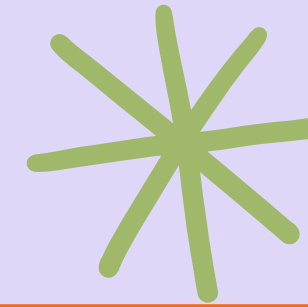
**NCERT Page Numbers**

# Quick Quiz

## Time!



Let's see what you remember! Think carefully and pick the right answer for each question below.



### Quiz 1



Does binary search work on an unsorted list? → No!

### Quiz 2



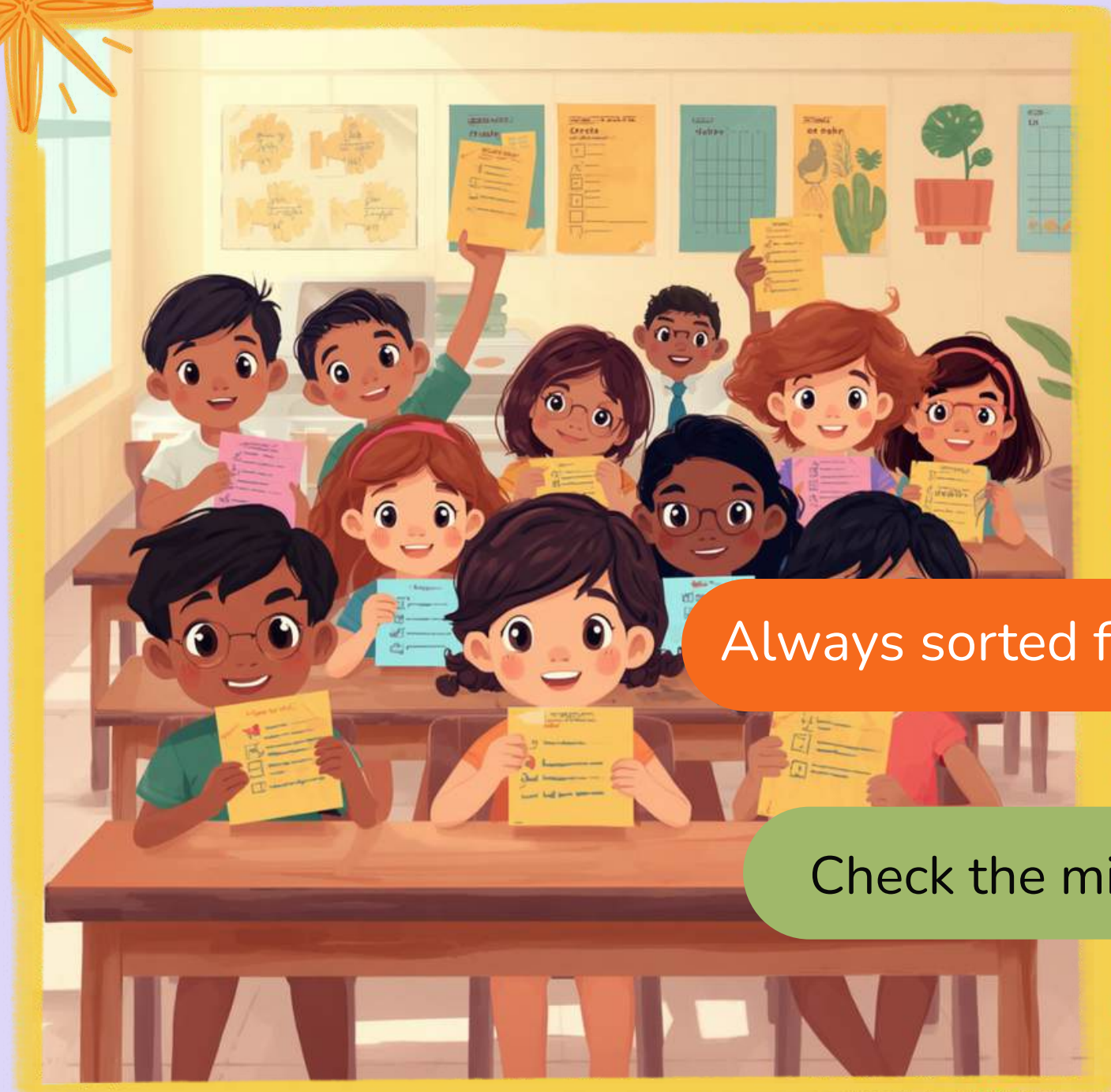
What do we do first in binary search? → Look in the middle!

### Quiz 3



Which search checks every item one by one? → Linear search!





Always sorted first!

Check the middle!

# Let's Remember!

**50%** Less time to find! Binary search halves the list each time!

Binary search is fast because it halves the list each time. It only works on sorted lists — order matters! We always start by checking the middle item. The left or right half is chosen based on our guess. Hands-on practice helps us understand better!





**Thank You &  
Keep Exploring!**

